SmartFlow NLEX Map Tab

Presentation script · Geospatial Analytics chapter discussion · Kiarra and Hansy · about 10 minutes

> **Before class:** sign in, open **Map Comparison** in Light theme, make sure the backend is running, and zoom the browser so both maps fit the projector. Whoever isn't speaking operates the laptop.

KIARRA Part 1: The problem and two core techniques

about 4 min

ON SCREEN **Map Comparison open, whole corridor showing**

"Good [morning/afternoon], Sir/Ma'am and classmates. We are from the SmartFlow NLEX group. Today Hansy and I will show how our system applies geospatial analytics, using one page of our dashboard: the Map tab.

Our question is simple: where is NLEX congested right now, and where will it be congested in the next hours? Traffic always happens in a place: on one side of the road, near one exit, at one hour. So we have to analyse it as a map, not just as numbers.

Our study area is the NLEX corridor, from Balintawak at Km 12 to Sta. Ines at Km 87. There are 20 exits and two carriageways, northbound and southbound."

ON SCREEN **Point to the left map, then the right**

"On the left is the live map, built from Waze reports from the last 60 minutes. On the right is our model's forecast for the same corridor. Both show the same road at the same scale, so we can compare now against later."

ON SCREEN **Point at the "Live · updated N min ago" badge**

"This badge shows how fresh the live data is."

The spatial data

"The map combines three kinds of spatial data:

- **Points:** things at one location, like Waze reports of hazards and accidents, our 20 toll plazas, and 20 weather points, one per exit.
- **Lines:** Waze sends each traffic jam as a line with its length, speed and delay.
- **Stretches:** data described by kilometre rather than coordinates, like a maintenance closure from Km 14.5 to 19.8, southbound."

Technique 1: Buffering and map matching

ON SCREEN **Zoom in near Bocaue or Paso de Blas on the live map**

"Waze draws jams on its own road network, so they often land beside the expressway, on a service road or a ramp. If we drew them as they came, we would show traffic on the wrong road. So we do three things:

1. **Buffer:** we keep only features within about 200 metres of the NLEX centreline. Reports from roads beside it are dropped.
2. **Snap:** we project each jam onto the centreline, so it sits on the road we draw.
3. **Direction:** we read northbound or southbound from the road name, or from the report's heading when the name doesn't say.

That is why each carriageway has its own ribbon here, and why a northbound jam never hides a southbound one."

Technique 2: Linear referencing

ON SCREEN **Scroll down to the Corridor panel under the maps**

"Once a jam is on the centreline, we can describe its position as one number: its kilometre post. This is how road agencies describe roads, by distance along the road instead of latitude and longitude.

This Corridor panel is the result: the whole corridor straightened into one line, from Km 12 to Km 87. Northbound is the top lane and southbound the bottom, and each queue is drawn at its real start and end kilometre.

One detail: the official km posts on NLEX signs don't exactly match the distance measured along our map geometry. So we show the official posts to people, but do the calculations on the measured distance.

Hansy will continue with proximity analysis and forecasting."

HANSY Part 2: Proximity, forecasting, design and limits

about 5 min

Technique 3: Proximity analysis

ON SCREEN **Hover a red or amber jam chip on the live map**

"Thank you, Kiarra. An operator doesn't think in coordinates. 'This jam is at 14.7 degrees north' means nothing to them. So for every queue and report, we find the nearest exit along the road and say which side of it the problem is on.

This card says the queue's length, the delay, the speed, and where it starts: for example, '422 metres past Harbor Link Toll Plaza'. 'Before' or 'past' comes from the direction of travel. That is a spatial join to the nearest exit, plus the direction.

We also add everything up per exit and direction. That is where counts like '6 congested, 6 slow' on the Overview come from."

ON SCREEN **Hover a toll plaza label, then click a Waze report pin if one is showing**

"A plaza card describes a place. A report card describes a condition: its reliability, when it was first reported at this spot, and which way it faces."

If there are no reports right now: "There are no Waze reports on the corridor at the moment, so I'll skip this one."

Linked views

ON SCREEN **Use "Jump to exit…" and pick Paso de Blas or Bocaue**

"Both maps move together, so live and forecast are always compared on the same stretch."

ON SCREEN **Press Clear, then click a spot on a lane in the Corridor panel**

"Clicking anywhere on the corridor line also moves both maps to that exact kilometre."

Forecasting over space and time

ON SCREEN **Point at the forecast map and its model badge**

"The right map is a forecast that is both spatial and temporal. For each of the 20 exits, the model predicts the traffic state for every hour up to 168 hours ahead, which is one week. There are three states, based on

Waze jam speeds: Moving, Heavy at 10 to 20 km/h, and Severe under 10 km/h.

We test it honestly. The model is scored only on the last 12 days of data it never saw. And it's only used if it beats a simple baseline: the usual state for that exit at that hour."

ON SCREEN **Read the badge aloud**

"Right now the badge says [model name], [X]% accurate."

If it says "Exit-hour profile": "This hour, no machine-learning model beat the baseline, so the system shows the baseline and says so, rather than using a weaker model."

ON SCREEN **Corridor panel: click +3 h on the time row**

"Choosing an hour switches the corridor line to the forecast for that hour. The predicted queue lengths come from years of past Waze jams at that exit and hour, so a forecast queue has a realistic length instead of covering the whole road."

ON SCREEN **Click "Live" to go back**

Map design

ON SCREEN **Zoom in closely on one carriageway**

"A few design choices make the map readable:

- Plaza names are white labels, because they are places. Jams are red or amber chips, because they are conditions.
- The small dashes move in the direction of travel: fast on clear road, slow in a queue, still when traffic has stopped.
- When labels would overlap, they move aside on leader lines."

ON SCREEN **Click Expand on the forecast map, then click one exit row**

"The expanded view lists every exit grouped as Congested, Slow or Clear, worst first, with the model's chance for each. Clicking one zooms the map to it."

ON SCREEN **Close the expanded view**

Limits

"Our map has limits:

- **Waze is crowdsourced.** A quiet stretch may be flowing, or simply have no one reporting.
- **Waze only sends speed inside a jam.** So we can't measure how fast clear road is moving.
- **Our geometry is carriageway level, not lane level.** A one-lane closure and a full closure look the same on the ribbon; the hover card gives the detail.
- **Traffic is very regular.** So a simple baseline is hard to beat, and we show which one wins."

KIARRA Closing

about 30 sec

"To sum up, the Map tab uses four geospatial ideas:

- A **buffer** keeps only what is really on NLEX.
- **Snapping** puts each jam on its carriageway.
- **Linear referencing** describes every place as a kilometre along the road.
- **Proximity analysis** says where things are relative to the exit an operator knows.

The forecast extends the same map into the coming hours. Thank you, Sir/Ma'am and classmates. We're happy to take your questions."

Likely questions

- **Why not use Google Maps traffic?** "Its data can't be stored or analysed. Waze gives us records we can save, snap to the road and score."
- **How accurate is the forecast?** Read the badge, then add: "That's measured on the 12 days the model never saw."
- **Which map projection?** "The map is drawn in Web Mercator by Mapbox. We compute distances in metres at the corridor's latitude."

If something fails live: "The live feed isn't reachable right now," then describe the step and move on.

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